RegreLineal200626

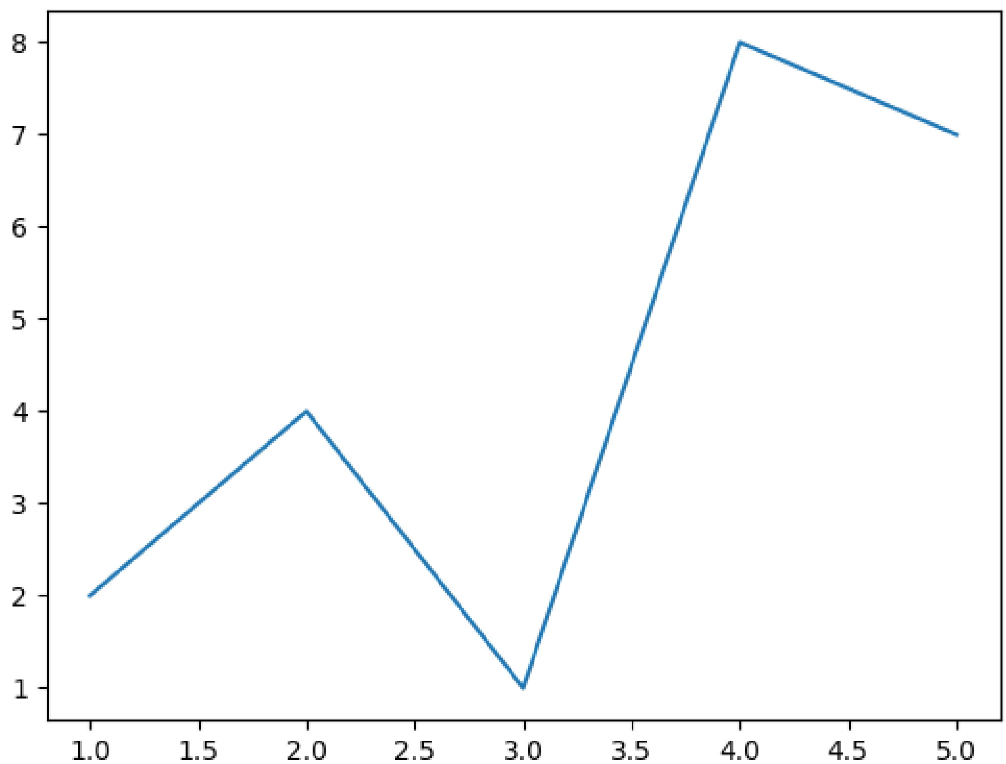
June 20, 2026

[2]:
```
# Importar librerias
import numpy as np
import pandas as pd
from sklearn import linear_model
import matplotlib.pyplot as plt
```

[3]:
```
x = [1,2,3,4,5]
y = [2,4,1,8,7]

plt.plot(x,y)
```

[3]: [<matplotlib.lines.Line2D at 0x17fd9ca90>]



[4]:
```python
# Para verificar si hay una relación lineal
correlacion = np.corrcoef(x,y)
correlacion
```

[4]:
```
array([[1.        , 0.72586619],
       [0.72586619, 1.        ]])
```

1 - Relación lineal positiva perfecta

0.8 - Muy fuerte

0.5 Moderada

0 no hay relacion lineal

-0.8 Relacion fuerte negativa

-1 Relacion negactiva perfecta

[5]:
```python
# Ejercicio ejemplo regresión lineal VENTAS
y = np.array([[7000],[9000],[5000],[11000],[10000],[13000]])
X = np.array([[1],[2],[3],[4],[5],[6]])
```

[6]:
```python
#Crear modelo vacio
regre_ventas = linear_model.LinearRegression()
```

[7]:
```python
regre_ventas.fit(X,y)
```

[7]:
```
LinearRegression()
```

[8]:
```python
# Ver la pendiente
regre_ventas.coef_
```

[8]:
```
array([[1114.28571429]])
```

[9]:
```python
# Ver la intersección de la linea
regre_ventas.intercept_
```

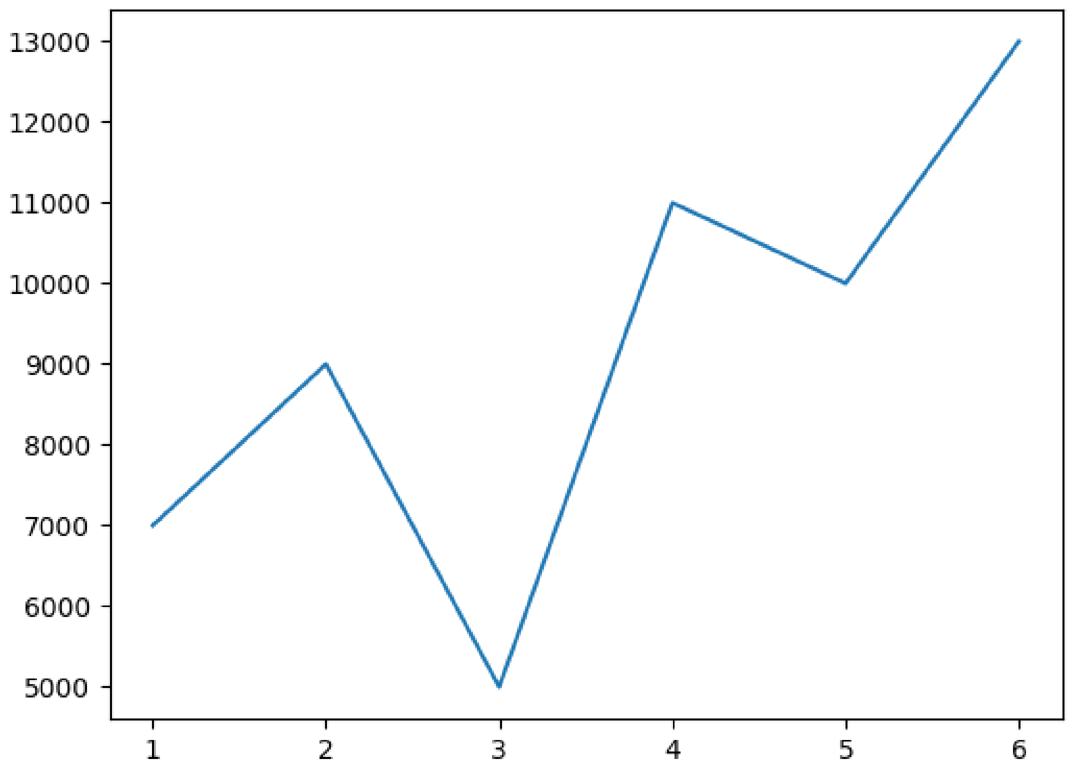
[9]:
```
array([5266.66666667])
```

[10]:
```python
# Voy a generar predicciones
regre_ventas.predict([[7]])
```

[10]:
```
array([[13066.66666667]])
```

[11]:
```python
# Graficar
plt.plot(X,y)
```

[11]:
```
[<matplotlib.lines.Line2D at 0x17feb5820>]
```



[16]:
```
y = [7000,9000,5000,11000,10000,13000]
X = [1,2,3,4,5,6]
```

[17]:
```
# Para verificar si hay una relación lineal
correlacion = np.corrcoef(X,y)
correlacion
```

[17]:
```
array([[1.        , 0.72947123],
       [0.72947123, 1.        ]])
```

[ ]:
```
```